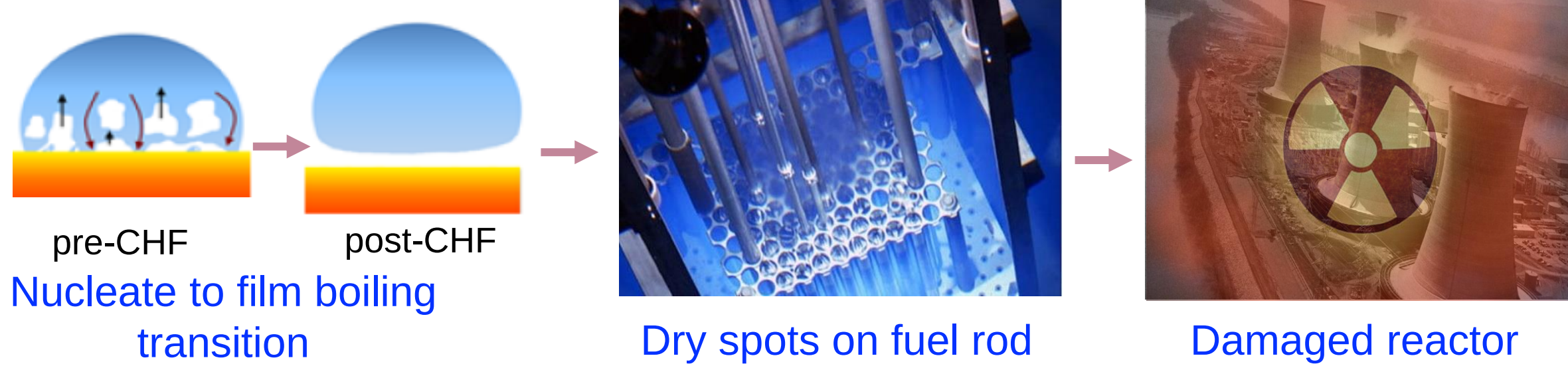
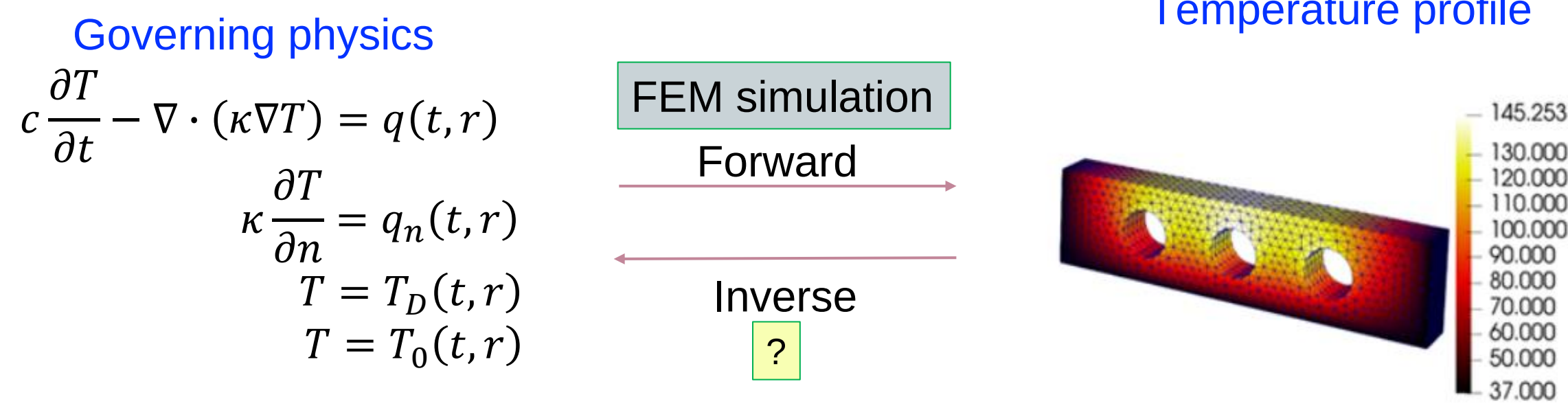


Heat transfer and safety in nuclear reactor



Predicting and preventing CHF requires solving IHCP

Inverse Heat Conduction Problem (IHCP)



- IHCP estimates unknown thermal parameters
- Heat flux is recovered by solving an IHCP
- Inverse problems require solving forward problems

Challenges in solving IHCP

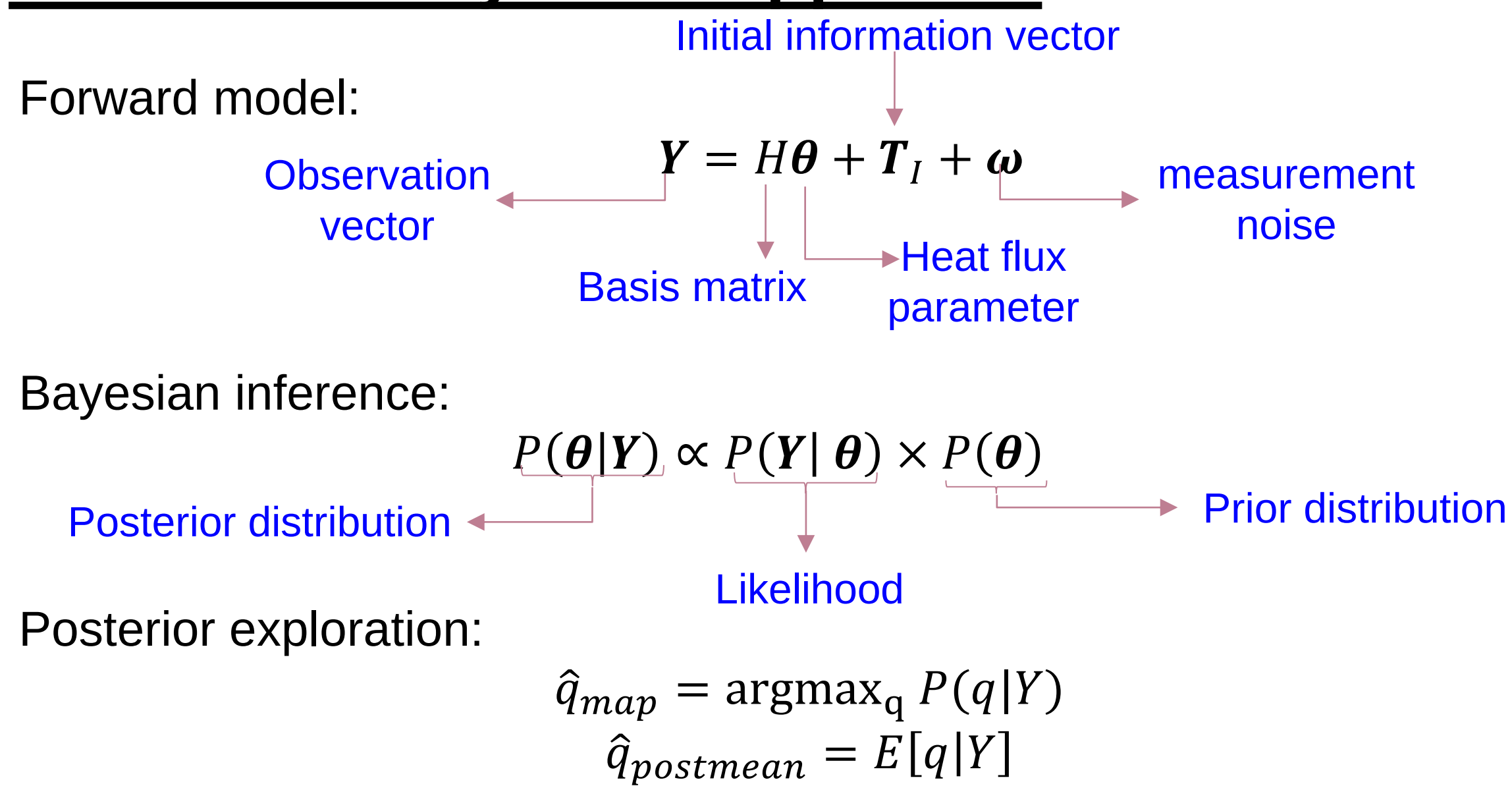
- Possibility of non-unique solutions
- Noise causes solution instability
- Forward problems are time-intensive

Limitations of Non-Bayesian Approach

- Unfit for non-linear scenarios
- Lack uncertainty quantification
- Highly sensitive to noise

The accurate quantification of uncertainty in recovered heat flux is crucial for ensuring reliable parameter retrieval.

Traditional Bayesian Approach



Limitations:

- Fails to consider spatio-temporal correlations in observed data
- Overlook important spatial and temporal correlations
- Solutions and predictions are less accurate

Observed data has spatio-temporal correlation that can be leveraged for improved recovery and prediction

Spatiotemporal Model

Input data → Spatial information + Temporal information + Interaction

This data type enables spatiotemporal modeling

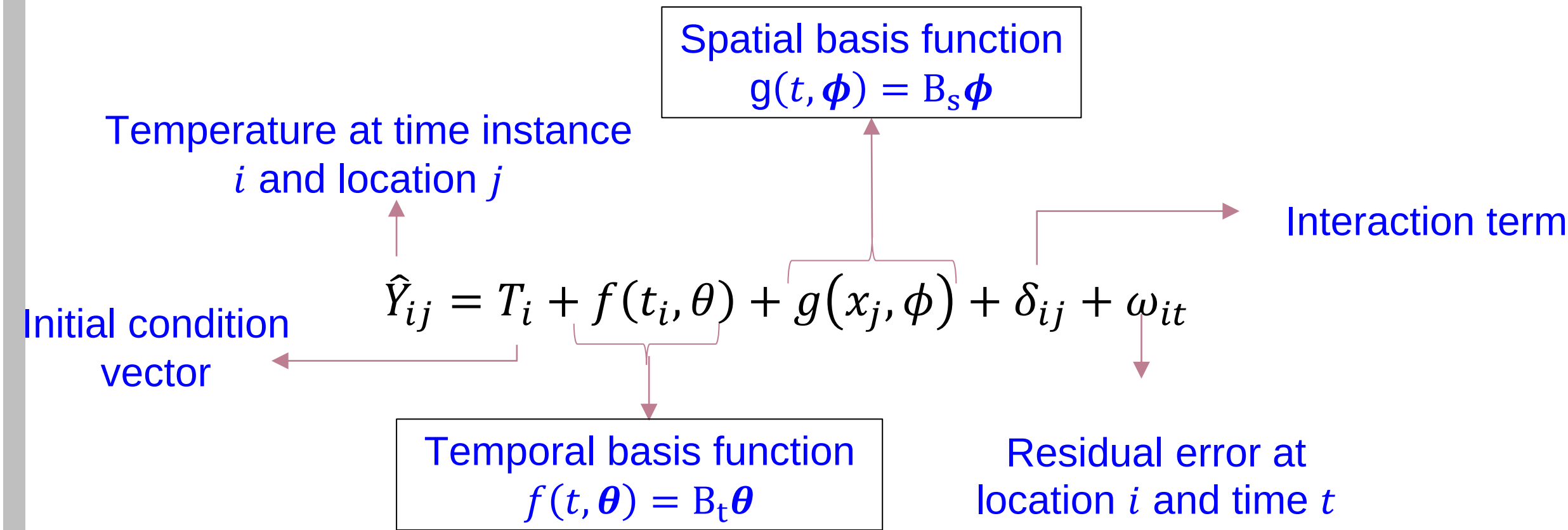
Step 1: Decompose temperature data into spatial and temporal components

$$T(t, r) = \sum_{i=1}^m \theta_i \omega_i(t) + \sum_{j=1}^p \phi_j v_j(r)$$

Temporal Components

Spatial Components

Step 2: Define the Spatiotemporal forward model



B_t and B_s are Basis matrices from predefine basis function

Step 3: Define the likelihood and prior model for Bayesian inference purpose

Spatial and temporal information is captured in the model through θ , ϕ , and δ .

Bayesian Inference for Spatiotemporal Model

$$P(\psi|Y) \propto P(Y|\psi) \cdot P(\psi)$$

Posterior distribution

Likelihood

Prior distribution

Lumped Parameter: $\psi = (\theta, \phi)$

$$\text{Likelihood: } P(Y|\theta, \phi) \propto \exp\left(\frac{(Y_{ij} - \hat{Y}_{ij})'(Y_{ij} - \hat{Y}_{ij})}{2\sigma^2}\right)$$

Priors:

$$P(\theta) \propto \exp\left(-\frac{1}{2} \lambda_1 \theta' W \theta\right); \quad P(\phi) \propto \exp\left(-\frac{1}{2} \lambda_2 \phi' V \phi\right)$$

- λ_1 and λ_2 are regularization parameters
- W and V are matrices of nonzero weights

Posterior

$$P(\theta, \phi|Y) \propto \exp\left(\frac{(Y_{ij} - \hat{Y}_{ij})'(Y_{ij} - \hat{Y}_{ij})}{2\sigma^2}\right) \times \exp\left(-\frac{1}{2} \lambda_1 \theta' W \theta\right) \times \exp\left(-\frac{1}{2} \lambda_2 \phi' V \phi\right)$$

Posterior Sampling: Gibbs Sampler

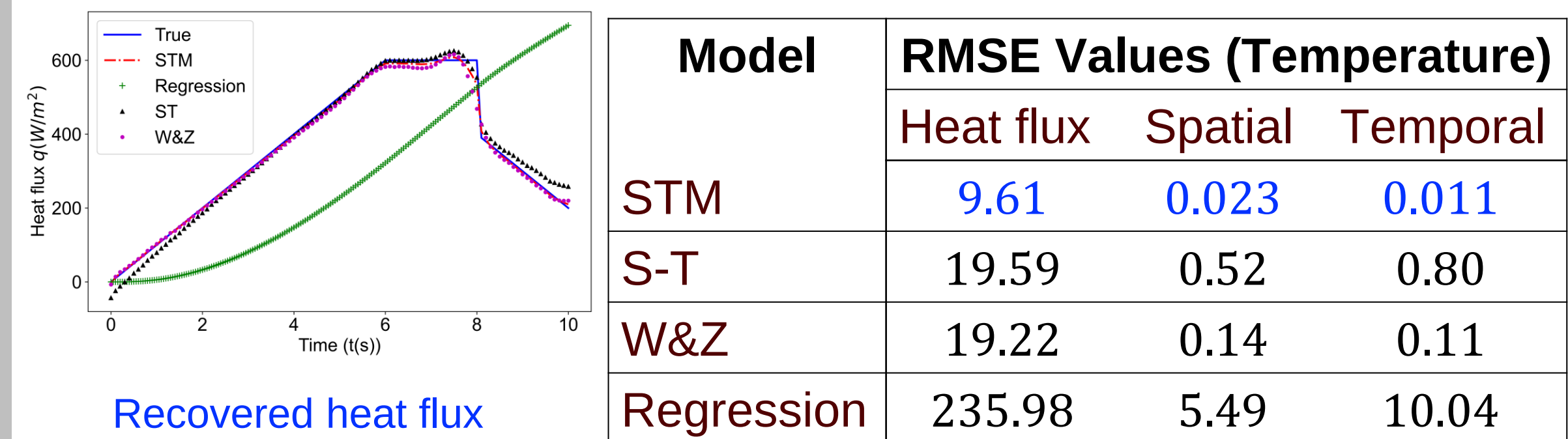
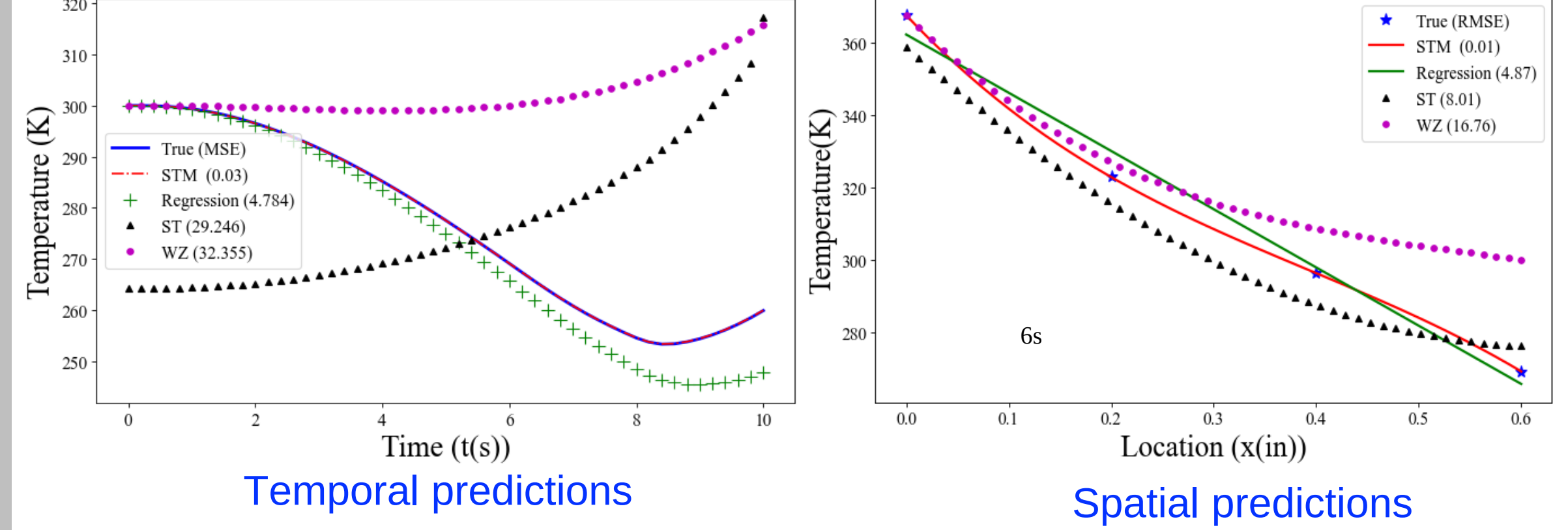
- Initialize $\theta^{(0)}, \phi^{(0)}$
- For $i = 0: N_{mcmc}$ (Total sampled points)
- Do:

$$\text{Sample } \theta_j^{(i+1)} \sim p\left(\theta_j^{(i+1)} \middle| \theta_{-j}^{(i)}, \phi^{(i)}\right) \quad \text{for } j \in [1, m]$$

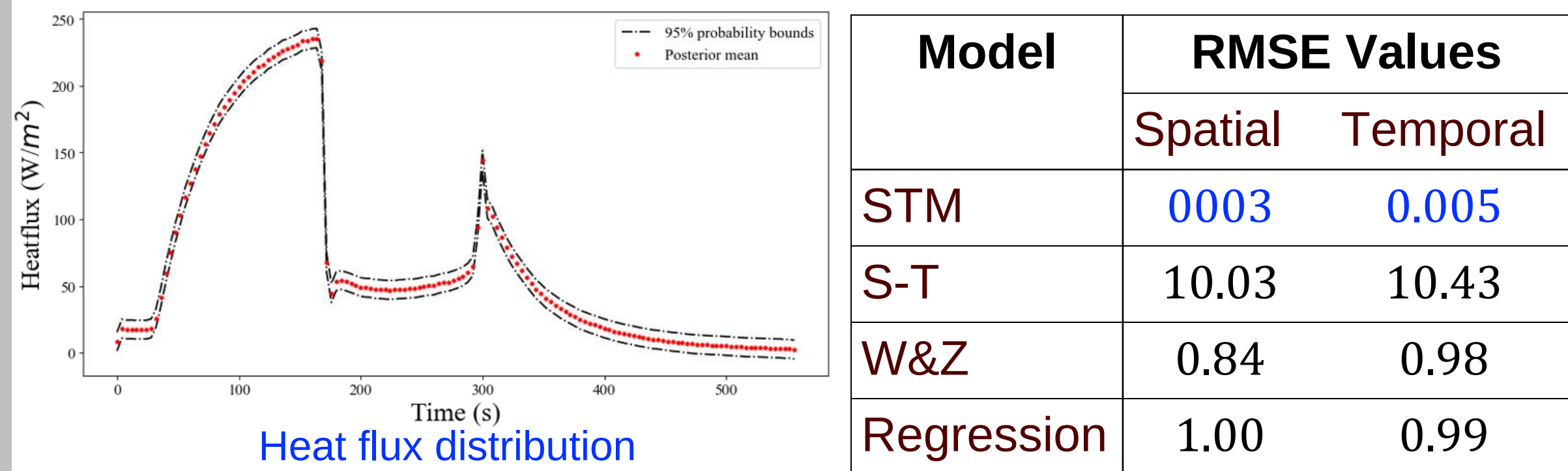
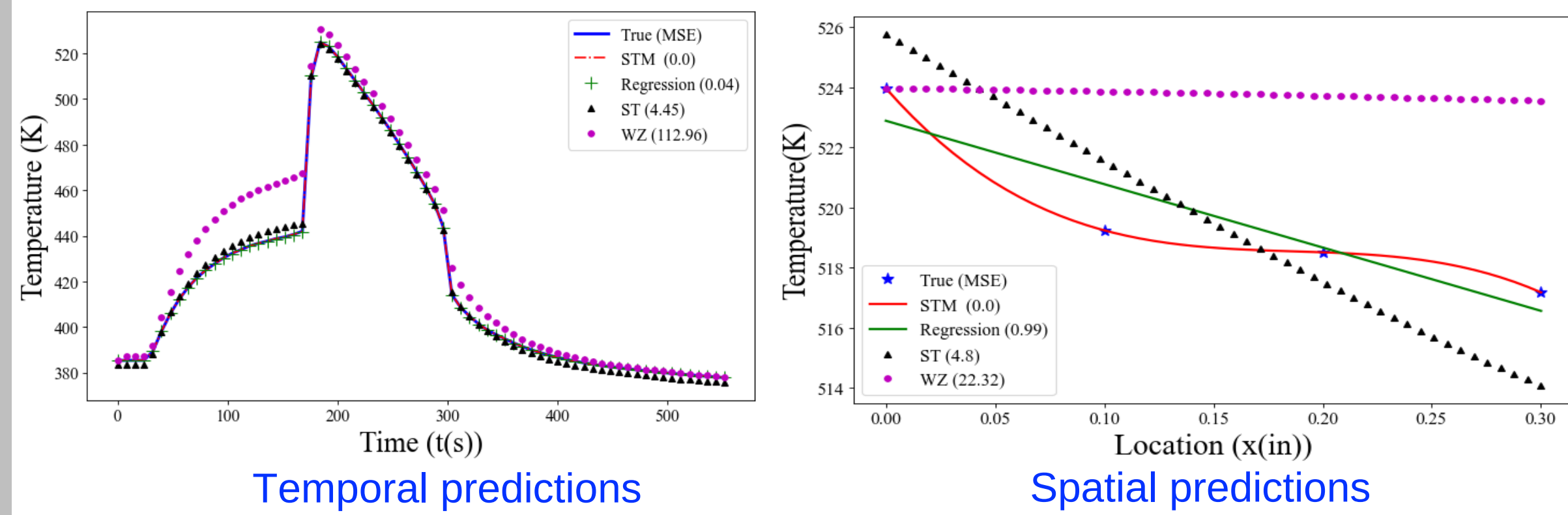
$$\text{Sample } \phi_k^{(i+1)} \sim p\left(\phi_k^{(i+1)} \middle| \phi_{-k}^{(i)}, \theta^{(i)}\right) \quad \text{for } k \in [1, p]$$

- Return sampled $\{\theta\}_1^{N_{mcmc}}$ and $\{\phi\}_1^{N_{mcmc}}$

Case Study (1D FEM simulations)



Case Study (Pool Boiling Experiment)



Summary

- Developed spatio-temporal modeling for heat flux recovery
- STM successfully recovered distributions from temperature data
- Proposed method performs better than competing method
- Interaction modeling improved performance

Next Step

- Hierarchical modeling of noise and parameters
- Multimodal data fusion for enhanced recovery

References

- Zayats, Galina, V. T. Borukhov, and G. M. Zayats. SSRN 4060732, 2022.
- Mohammad-Djafari, Ali. Entropy 23.12: 1673. 2021
- Wang, Jingbo and Zabarar, Nicholas. International Journal of Heat and Mass Transfer, 2004